

$$\dot{x} = f(x);$$

Find fixed points, analyze their stability, plot $x(t)$, phase portrait, plot $\dot{x}$ vs x

$$\dot{x}(t) = f(x(t))$$

① Find x^* such that $f(x^*) = 0$.

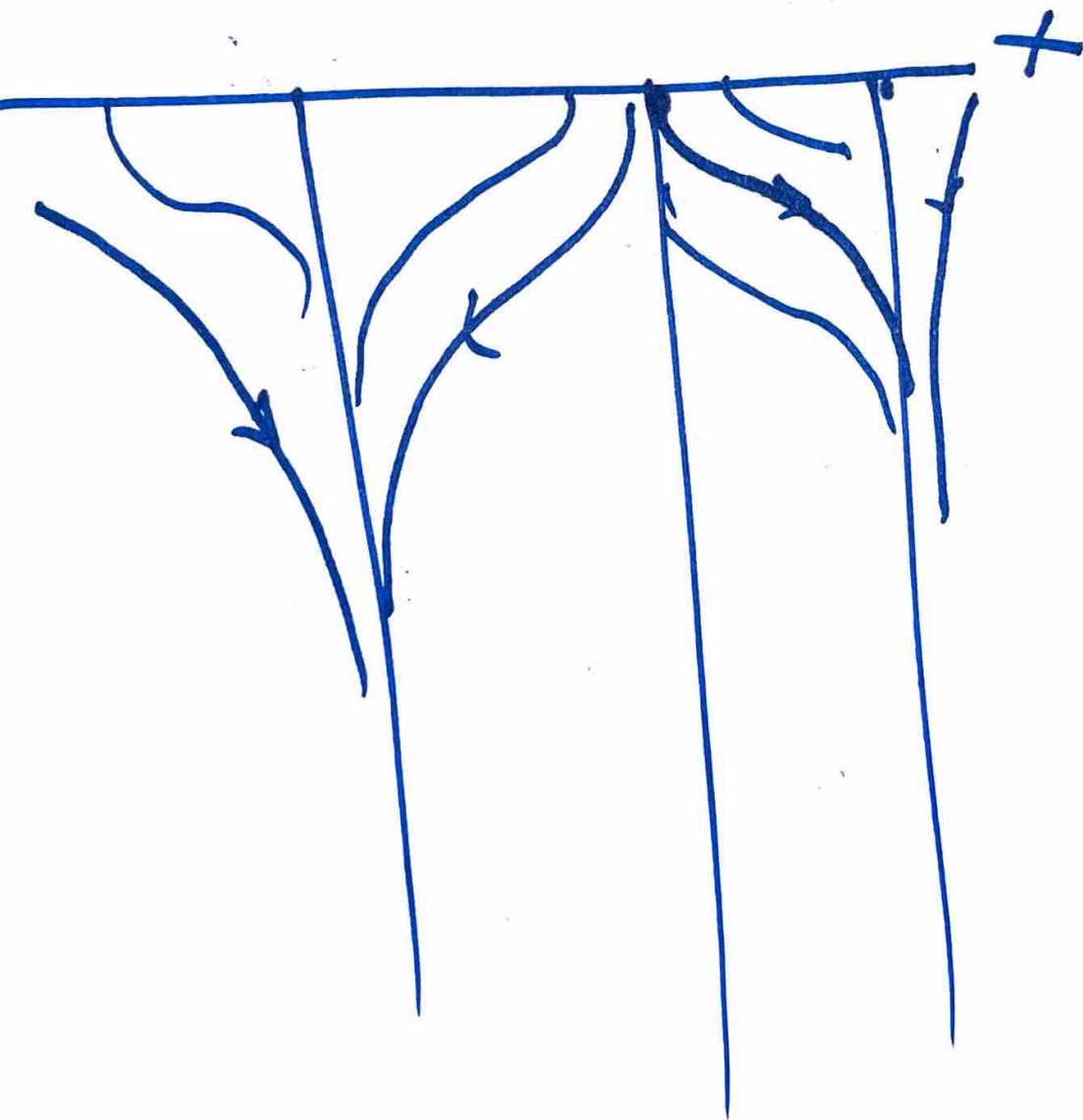
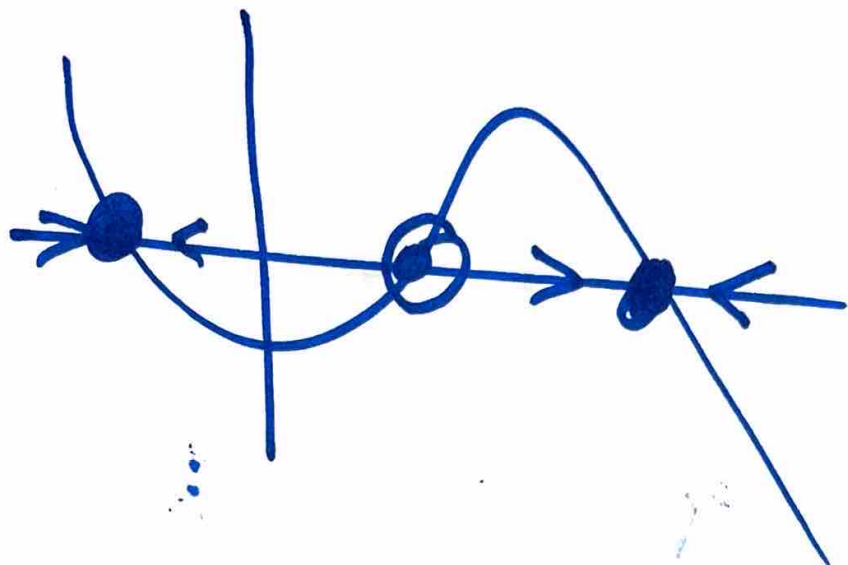
② analyze $f(x)$ in between fixed points

(a) $f(x) > 0$, $\dot{x} > 0$, $x(t)$ grows, motion to the right

(b) $f(x) < 0$, $\dot{x} < 0$, $x(t)$ decays, motion to the left

(c) $f(x) = 0$ - fixed point, do not move

3



t

-4-

$$\dot{x} = f(x)$$

x^* - fixed

$$x = x^* \quad x(t) = x^* + \eta(t) + \dots + \eta(t)$$

$$\dot{x}^* + \dot{\eta}(t) = f(x^* + \eta(t))$$

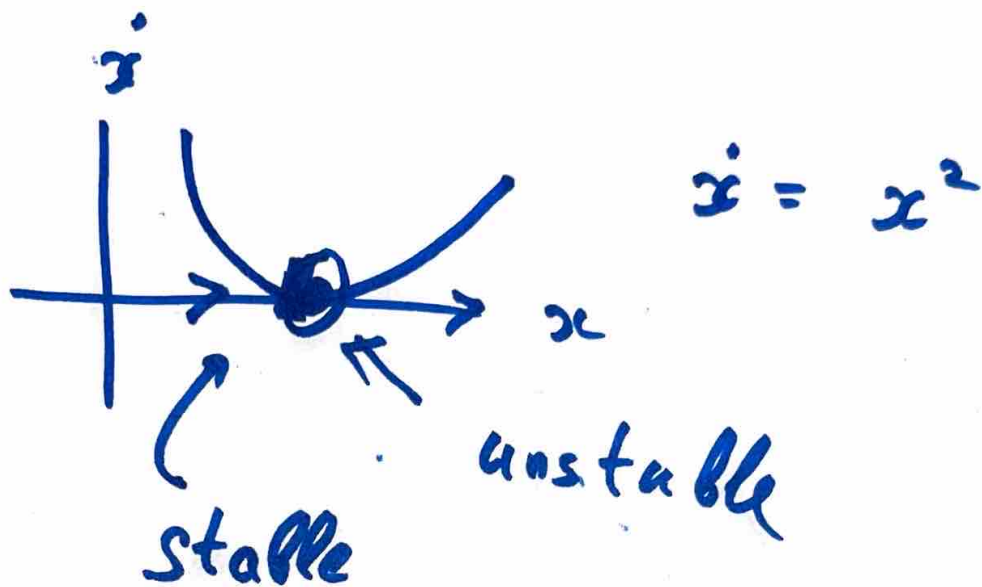
$$\approx f(x^*) + f'(x^*) \eta(t) + \dots$$

$$\dot{\eta}(t) = f'(x^*) \eta(t)$$

IF $f'(x^*) > 0$ - unstable

$f'(x^*) < 0$ stable

-5'



$$\dot{x} = - \frac{\partial V(x(t))}{\partial x}$$

$$V(x) = - \int_0^x f(x) dx$$

$$\frac{d}{dt}$$

$$\frac{d}{dt} V(x(t)) \leq 0$$

-5'

$$\dot{x} = x(x^2 - r)$$

$$\dot{x} = x(x^2 + r)$$

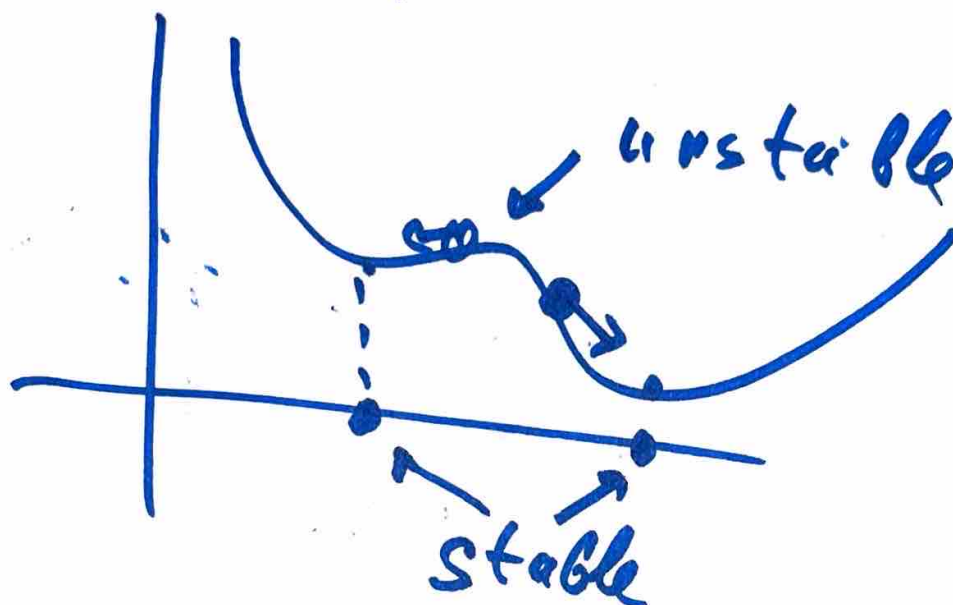
" mod. fixed pitch for A)

$$\dot{x} = x(x^2 + r) - x^5$$

1
-

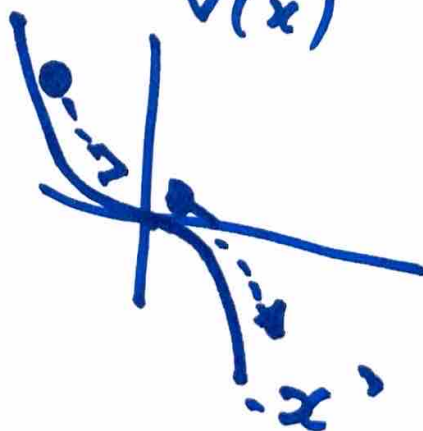
-6-

$V(x)$



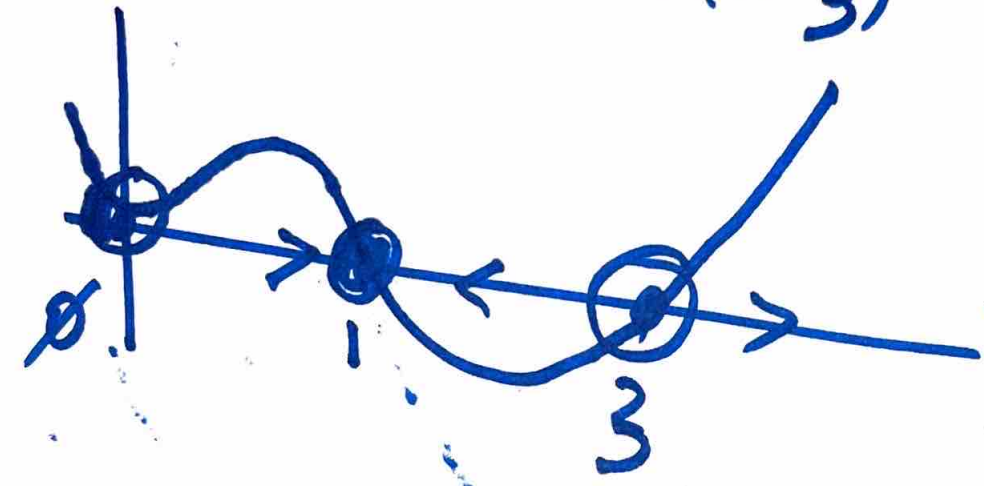
$$\dot{x} = x^2 = -\frac{d}{dx} V(x)$$

$$V(x) = -\int x^2 dx = -\frac{x^3}{3}$$



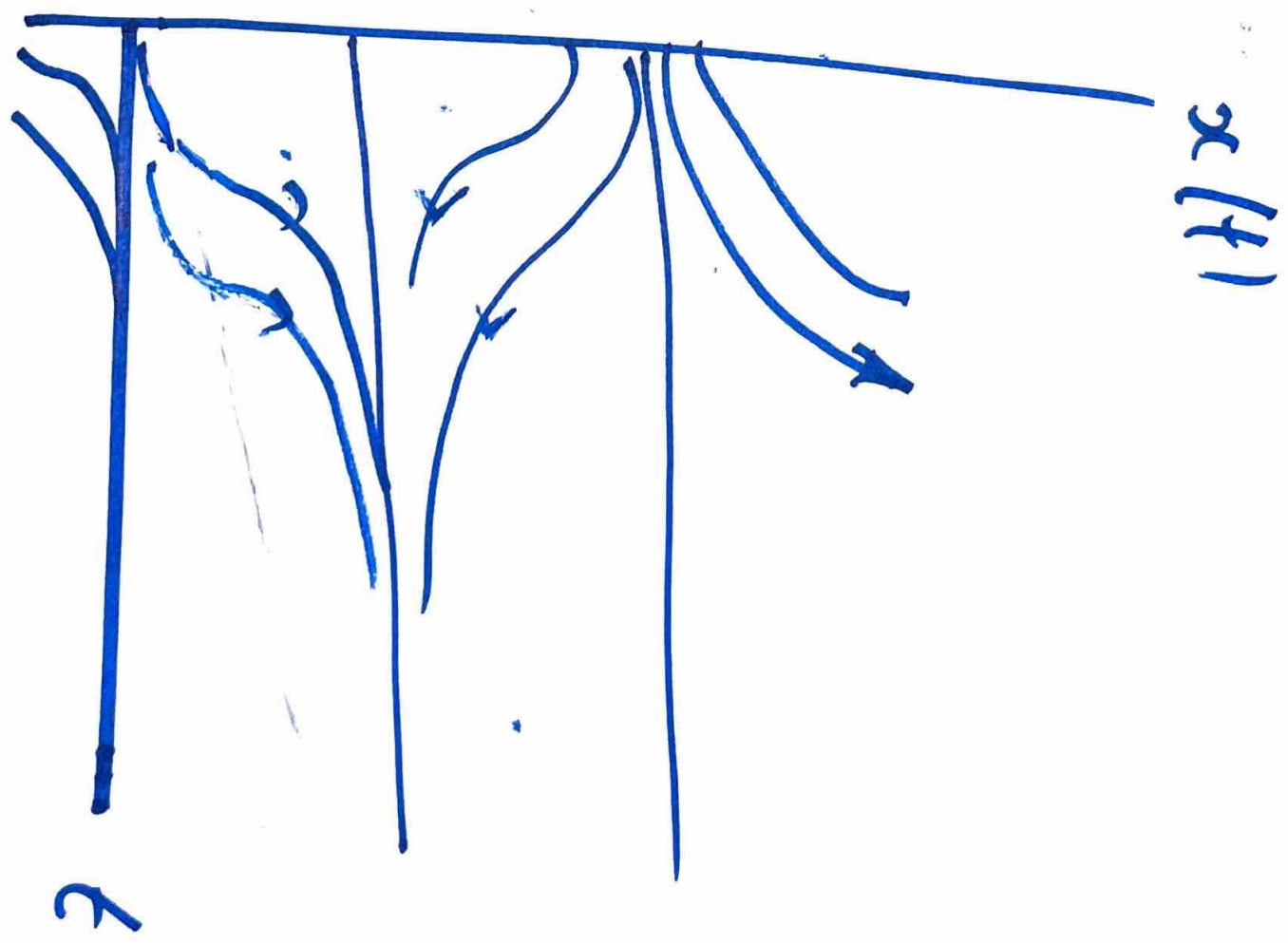
-t-

$$\dot{x} = x^2 (x-1)(x-3)$$



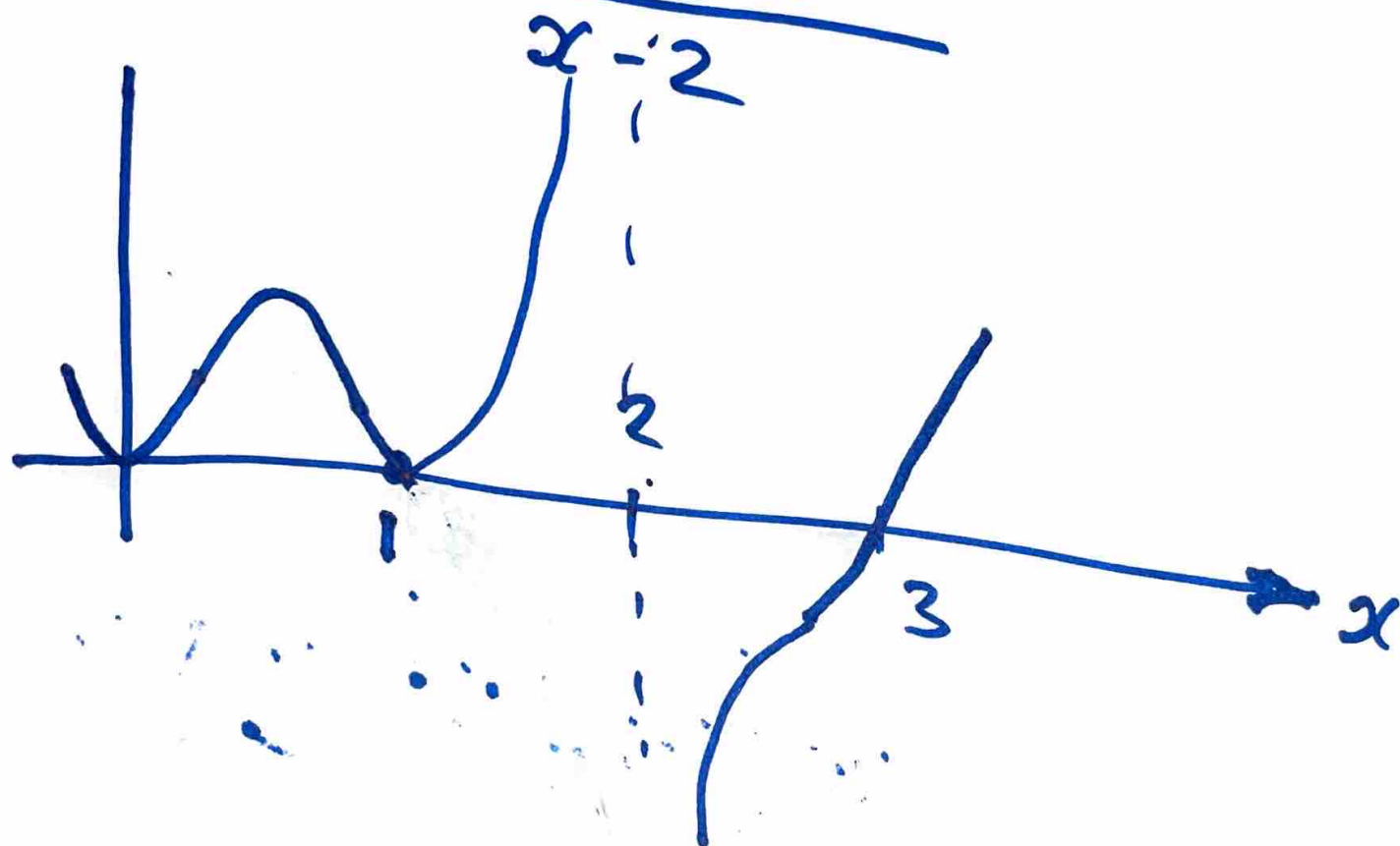
$x^* = 1$ stable
 $x^* = 3$ unstable
 $x^* = 0$

mixed
stability



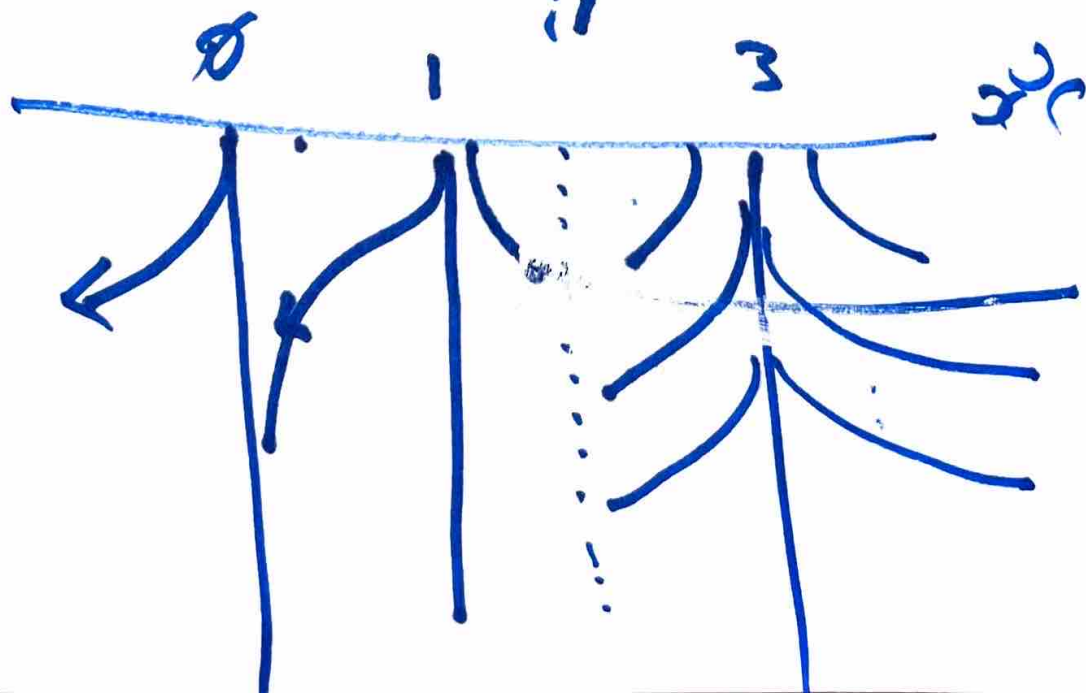
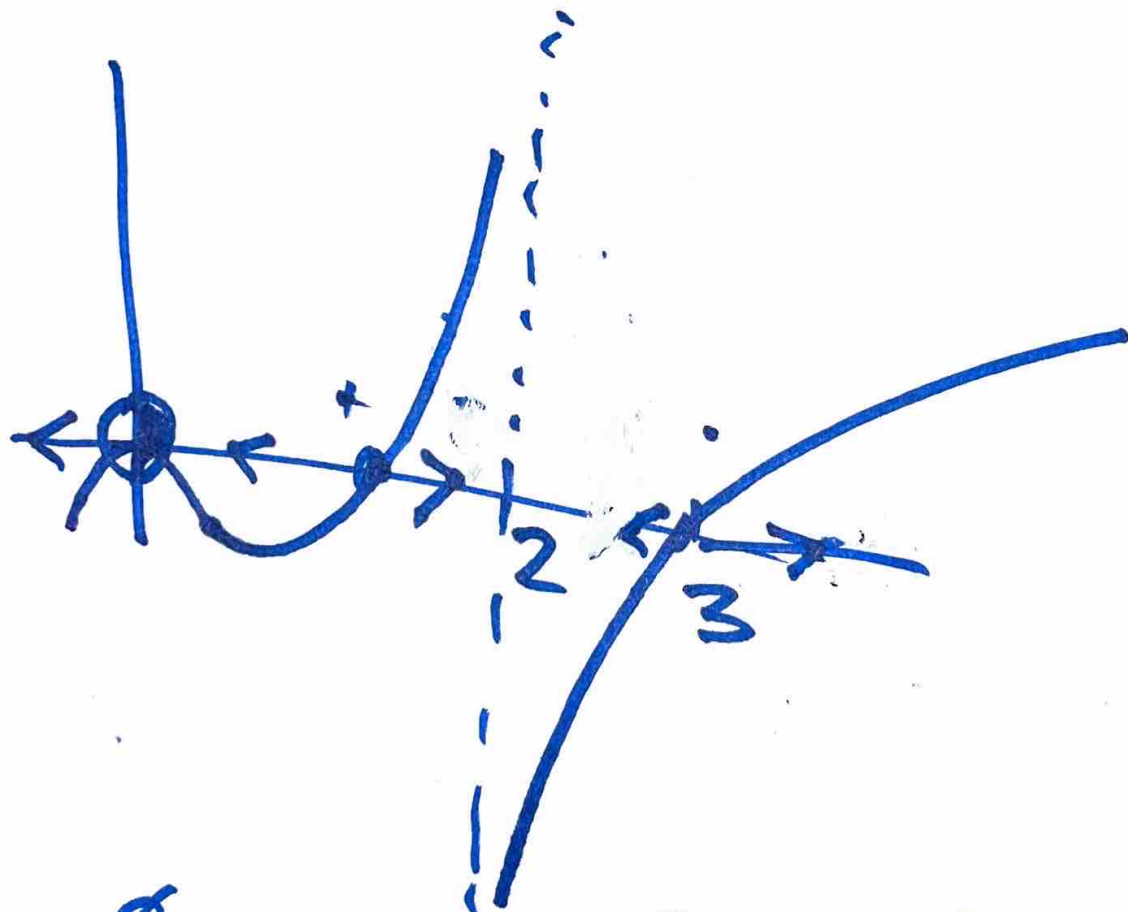
-8-

$$\dot{x} = \frac{x^2(x-1)(x-3)}{x-2}$$



-8

$$\dot{x} = \frac{x^2(x-1)(x-3)}{x-2}$$

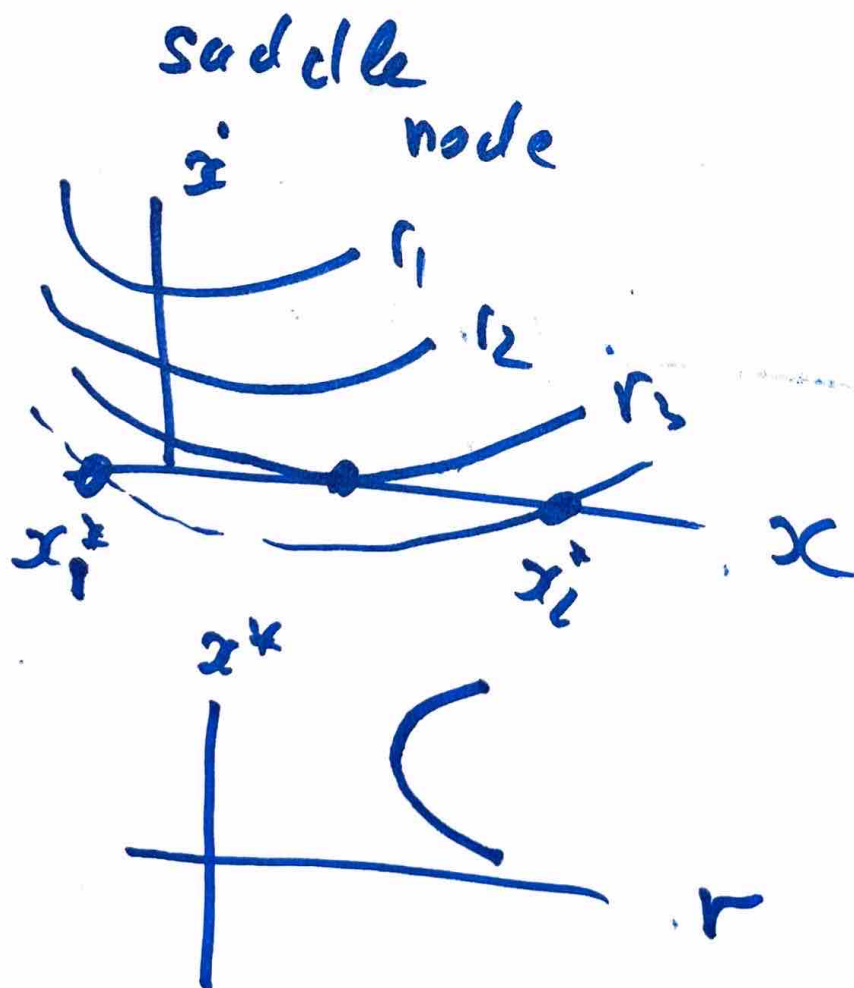


9-

Bifurcations

$$\dot{x}(t) = f(x(t), r)$$

$$\dot{x} = r + x^2$$



-9-

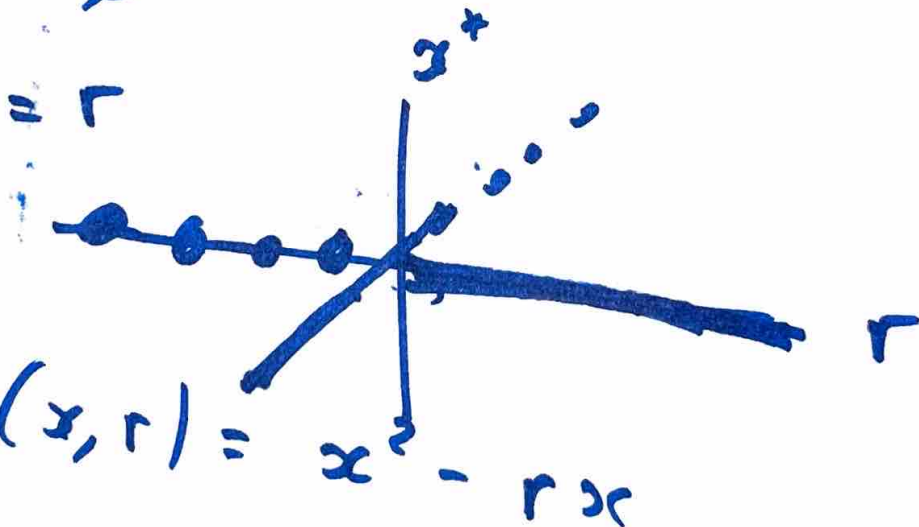
Transcritical

method

$$\dot{x} = x(x-r)$$

$$x^* = 0$$

$$x^* = r$$



$$\dot{x} = f(x, r) = x^2 - rx$$

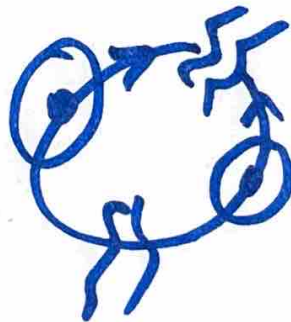
$$\frac{\partial f}{\partial x} = 2x - r$$

IF $x^* = 0$, stable $r > 0$
unstable $r < 0$

$x^* = r$, unstable $r > 0$
stable $r < 0$

9-

$\dot{\theta} =$ periodic



-10-

$$\dot{x}(t) = f(x) = g(x, r)$$

fixed point

$$f(x^*) = g(x^*, r)$$

$$\frac{df}{dx} = \frac{\partial g(x, r)}{\partial x}$$

- 11 -

$$\frac{d}{dt} \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$$

$$x(t) = C_1 \underline{x}_1 e^{\lambda_1 t} + C_2 \underline{x}_2 e^{\lambda_2 t}$$

$$\text{IF } \lambda_1 \neq \lambda_2 \\ \underline{x}_1 \neq \underline{x}_2$$

$$\underline{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

$$\text{Det} [\underline{A} - \lambda \underline{I}] = 0$$

$$\text{Det} \left[\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right] = 0$$

$$\dot{\underline{x}} = \underline{A} \underline{x}$$

-12 -

$$(a_{11} - \lambda)(a_{22} - \lambda) - a_{12}a_{21} = 0$$

$$(\lambda - a_{11})(\lambda - a_{22}) - a_{12}a_{21} = 0$$

$$\lambda^2 - (a_{11} + a_{22})\lambda + a_{11}a_{22} - a_{12}a_{21} = 0$$

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

$$a_{11} + a_{22} = \text{Tr}(A) - \text{trace}$$

$$a_{11}a_{22} - a_{12}a_{21} = \text{Det}(A) \\ \text{determinant}$$

-13-

$$\lambda^2 - \lambda \cdot \text{Tr}(A) + \text{Det}(A) = 0$$

$$\tau \equiv \text{Trace}(A) = a_{11} + a_{22}$$

$$\Delta \equiv \text{Det}(A) = a_{11}a_{22} - a_{12}a_{21}$$

$$\lambda^2 - \tau\lambda + \Delta = 0$$

$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}$$

$$\tau^2 > 4\Delta > 0$$

$$\Delta > 0, \tau > 0$$

$$\sqrt{\tau^2 - 4\Delta} < \tau$$

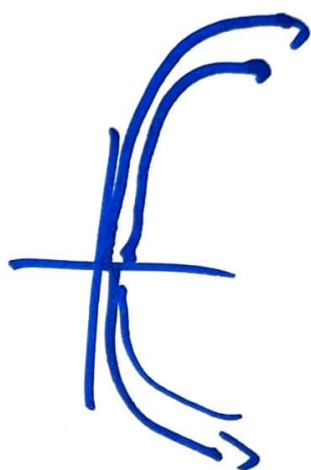
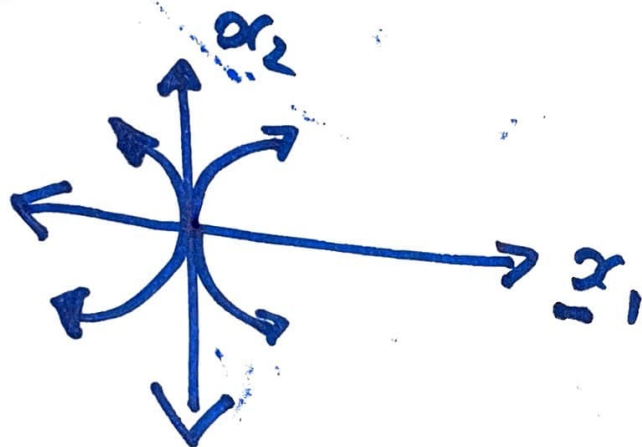
$$\tau^2 - 4\Delta < \tau^2$$

$$-4\Delta < 0$$

-14-

$$T^2 > 4\Delta; \Delta > 0; T > 0;$$

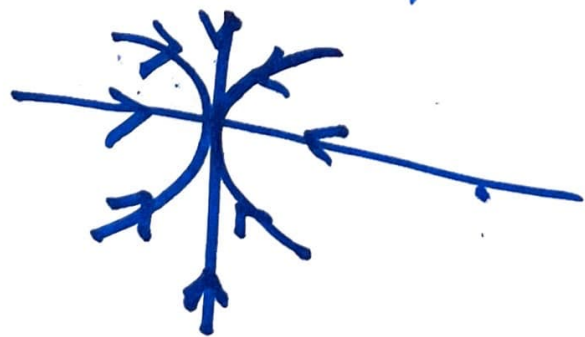
$$\lambda_1 > \lambda_2 > 0$$



Unstable
node

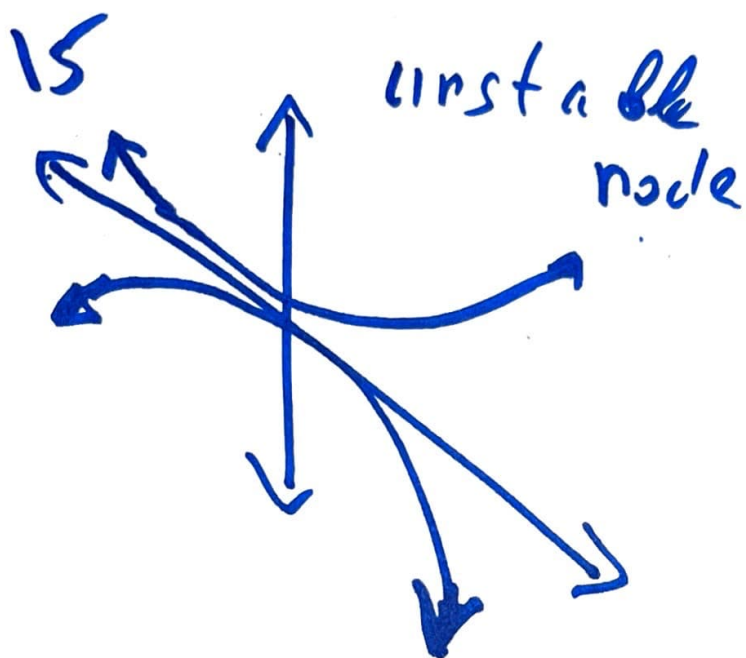
$$T^2 > 4\Delta; \Delta > 0; T < 0.$$

$$\lambda_1 < \lambda_2 < 0$$



Stable
node





$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}$$

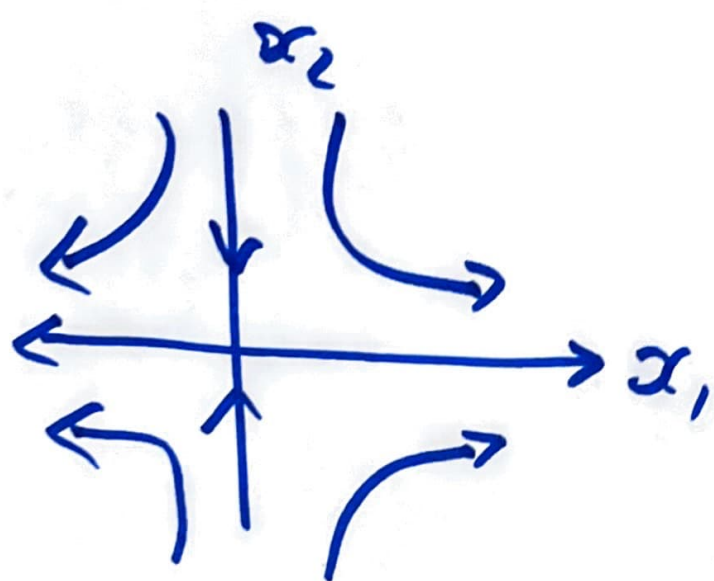
$$\tau^2 > 4\Delta,$$

$$\Delta < 0$$

$$\sqrt{\tau^2 - 4\Delta} > |\tau|$$

$$\lambda_1 < 0, \lambda_2 > 0 \quad \lambda_1 < 0 < \lambda_2$$

16



Saddle
point